

Benchmark paper

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Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility.

We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models.

It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. Standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI.

1 Introduction

Diffusion magnetic resonance imaging (dMRI) enables the in vivo quantification of tissue microstructure and provides biologically meaningful biomarkers sensitive to axonal organization, myelination, and tissue complexity [10]. Leveraging these microstructural descriptors for predictive modeling has become increasingly common, with applications spanning demographic estimation [2, 7], clinical diagnosis [6, 15], and cognitive characterization [9]. The combination of diffusion-derived features and machine learning is therefore often viewed as a powerful framework for data-driven neurobiological inference.

Yet despite rapid methodological development and steadily increasing reported accuracies, a fundamental question remains unresolved: how robust are these gains? Reported performance varies widely across studies, tasks, and datasets, and improvements are frequently attributed to architectural innovation or feature engineering. However, direct comparison between studies is rarely possible, and it remains unclear whether observed gains reflect genuine microstructural signal, dataset-specific properties, or sensitivity to methodological choices. In the absence of standardized evaluation, diffusion MRI prediction risks becoming a comparison of configurations rather than a cumulative scientific endeavor.

Several recurrent sources of inconsistency contribute to this fragmentation. Preprocessing pipelines and quality-control procedures differ substantially across cohorts. Microstructural information is represented through heterogeneous strategies (e.g., voxel-wise maps, ROI summaries, or learned embeddings), yet evaluations rarely disentangle representation effects from modeling effects. Evaluation protocols vary in split definitions, cross-validation schemes, and hyperparameter selection, often conflating methodological flexibility with performance gains. Finally, incomplete release of code and configuration details further limits reproducibility. Together, these factors make it difficult to determine which reported improvements are stable, generalizable, and biologically meaningful.

To address these limitations, we introduce a standardized, open-source diffusion MRI benchmark that enables controlled, leakage-free comparison across datasets, tissue compartments, microstructural representations, and model families. Rather than focusing solely on peak accuracy, the benchmark explicitly quantifies performance variability across folds and configurations, isolates representation and tissue-specific effects, and evaluates the incremental value of diffusion-derived features beyond non-diffusion baselines. By unifying preprocessing, feature extraction, model selection, and evaluation under a single experimental framework, this work aims to shift diffusion MRI prediction from isolated performance claims toward variance-aware, reproducible benchmarking.

Our contributions are fourfold: (i) a standardized preprocessing and data preparation pipeline; (ii) a unified feature-extraction framework supporting multiple microstructural representations for gray and white matter; (iii) a consistent evaluation protocol with leakage-free cross-validation and configuration-level statistical analysis; and (iv) the public release of code, splits, and experimental configurations to enable transparent reuse and extension. Collectively, this benchmark provides the foundation for disentangling genuine microstructural signal from configuration-driven variability and for establishing comparable evaluation standards in diffusion MRI prediction.

2 Benchmark

Our proposed benchmark provides a standardized, end-to-end framework for evaluating predictive models on diffusion MRI data. As illustrated in Figure 1, the pipeline comprises several interconnected modules: uniform preprocessing, microstructural feature extraction, predictive modeling, and rigorous evaluation. Specifically, our benchmark applies a unified preprocessing pipeline to all cohorts—with the exception of the HCP dataset, which utilizes its own established preprocessing pipeline. Following preprocessing, we perform tailored feature extraction to capture tissue-specific microstructural representations. Finally, we apply a rigorous, standardized cross-validation and analysis protocol to evaluate a wide spectrum of predictive models, ensuring reproducibility and fair comparison.

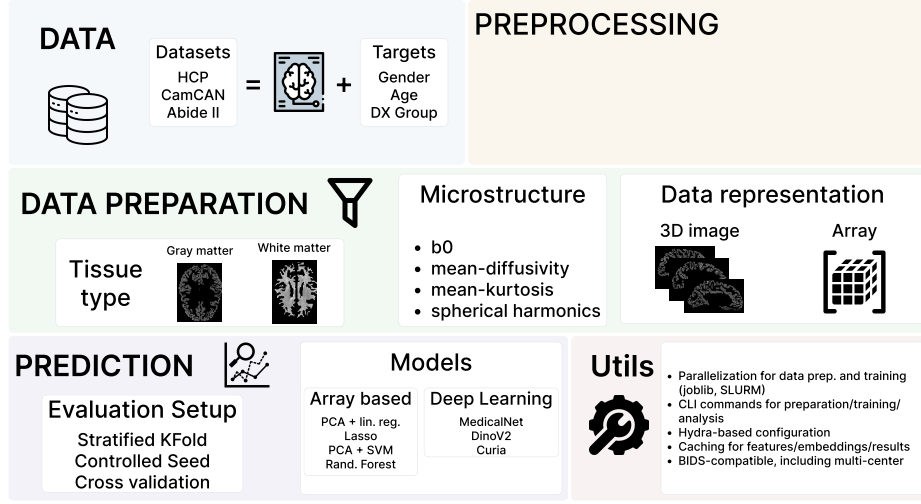


Fig. 1. Overview of the proposed diffusion MRI benchmark pipeline, illustrating the end-to-end process from preprocessing to microstructural feature extraction, predictive modeling, and evaluation.

2.1 Datasets

To ensure reproducibility, we evaluate our benchmark on three publicly available datasets covering both healthy and clinical populations, offering diverse demographics, scanner types, and acquisition protocols. For healthy cohorts, we utilize the Human Connectome Project (HCP) Young Adult dataset [14] and the Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset [12], both targeting gender and age prediction (classification and regression, respectively). After filtering, the HCP cohort comprises 1,036 subjects (474 males, 562 females; ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b -values = 1000, 2000, 3000 s/mm², 90 directions/shell). The Cam-CAN cohort includes 629 subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b -values = 1000, 2000 s/mm², 30 directions each).

For clinical evaluation, we use the Autism Brain Imaging Data Exchange II (ABIDE II) dataset [5], chosen over ABIDE I due to the availability of diffusion MRI. Targeting Autism Spectrum Disorder (ASD) diagnosis (classification), the filtered ABIDE II cohort consists of 218 subjects (186 males, 32 females; ages 5-64), including 119 with ASD and 99 controls. As a multi-site dataset, ABIDE II utilizes various 3T scanners (e.g., Siemens, GE) with highly heterogeneous, site-specific single-shell diffusion protocols. Across the included sites, spatial resolutions range from 0.94 to 3.0 mm, TR/TE from 5200-20244/78-101 ms, b -values include 1000, 1500, and 2500 s/mm², and diffusion directions vary between 32

and 64. This extreme variability provides a realistic scenario for evaluating model robustness to scanner heterogeneity.

2.2 Preprocessing and Feature Extraction

Preprocessing (Himanshu)

- Denoising
- Motion and eddy correction
- Spatial normalization
- Masking strategy

Microstructural Representation

Gray Matter Representation Gray matter microstructure was represented on the subject-specific cortical surface reconstructed from **T1-weighted images using the Desikan–Killiany atlas (aparc)**. For each subject, cortical meshes (white/pial surfaces) were generated and diffusion-derived scalar maps were computed in native diffusion space and rigidly aligned to the anatomical space. Voxelwise diffusion measures were then projected onto the individual cortical surface using ribbon-constrained sampling, preserving vertex-wise resolution (64k vertices per hemisphere). This subject-specific surface representation was chosen to respect cortical geometry, reduce partial-volume effects, and maintain high spatial specificity while enabling anatomically consistent regional correspondence via the atlas labels across subjects.

White Matter Representation White matter microstructure was represented within a standardized tract space using the Tract-Based Spatial Statistics (TBSS) framework [13]. Diffusion-derived scalar maps were nonlinearly registered to standard space, projected onto the group-wise white matter skeleton to minimize partial volume effects and residual misalignment, and summarized using the JHU White-Matter Tractography Atlas. Mean microstructural values were extracted from 48 major bundles defined in the atlas, yielding a compact tract-level representation per metric. We intentionally adopted this skeleton-based, atlas-constrained strategy rather than subject-specific tractography to enforce anatomical correspondence across subjects, reduce variability introduced by tracking parameters, and enhance robustness in a multi-dataset benchmarking setting. Although this representation sacrifices voxel-level granularity relative to the cortical surface analysis, it provides reproducible, homologous bundle definitions across datasets and mitigates registration and tract delineation inconsistencies—thereby strengthening cross-cohort comparability and methodological stability.

Microstructural models and normalization To comprehensively capture the biological properties of brain tissue, we compute a diverse set of microstructural metrics using established diffusion models. These metrics were selected to represent different facets of tissue architecture, ranging from basic diffusion magnitude to complex restricted environments:

- **Baseline ($b = 0$):** The averaged T2-weighted non-diffusion signal is included as a baseline to disentangle purely diffusion-driven predictive power from underlying structural T2 contrast.
- **Mean Diffusivity (MD):** Computed via Diffusion Tensor Imaging (DTI) [1], representing the overall magnitude of water diffusion.
- **Mean Kurtosis (MK):** Derived from Diffusion Kurtosis Imaging (DKI) [8], capturing the non-Gaussianity of diffusion as a proxy for microstructural heterogeneity and tissue complexity.
- **Spherical Harmonics (SH) Power:** A model-free representation obtained by computing the L2 norm of the SH coefficients (up to order 6) directly from the diffusion signal, capturing signal complexity and anisotropy without assuming a specific tissue compartment model.

We apply a robust ventricular normalization strategy. For each subject, voxel-wise diffusion metrics are divided by the mean metric value extracted from the cerebrospinal fluid (ventricles). Finally, extreme pathological values are clipped at the 99th percentile to ensure numerical stability during model training.

2.3 Prediction models

To establish a comprehensive evaluation, our benchmark incorporates a wide spectrum of predictive models, ranging from simple baselines to advanced deep learning architectures. First, we include dummy estimators (majority class for classification and mean predictor for regression) to provide a lower-bound performance baseline. For classical machine learning, we evaluate linear models (e.g., logistic regression, Lasso), Random Forests, and Support Vector Machines (SVM). Given the high dimensionality of the microstructural representations, we also evaluate these models coupled with Principal Component Analysis (PCA) for feature reduction.

For deep learning, we benchmark several vision architectures. We leverage pre-trained foundation models, specifically DINOv2 [11] and Curia [4], utilizing them as frozen feature extractors where representations are computed offline prior to training the prediction head. In contrast, 3D architectures like MedicalNet [3] are fully fine-tuned on our diffusion datasets to adapt to the specific microstructural domains. **To ensure a fair and rigorous comparison, all models are optimized using a unified hyperparameter search strategy (e.g., grid or random search) with internal cross-validation strictly confined to the training set to prevent data leakage.**

2.4 Benchmarks tasks and evaluation metrics

The selection of prediction tasks in our benchmark is driven by clinical relevance, data availability, and the need to align with established benchmarks in the neuroimaging literature. We evaluate models across two primary paradigms:

Classification For binary classification, we focus on tasks such as Autism Spectrum Disorder (ASD) diagnosis (ASD vs. Typically Developing) using the ABIDE II dataset, as well as gender prediction in the HCP and Cam-CAN cohorts. For all classification tasks, binary targets are strictly encoded as 0 and 1. To comprehensively assess classification performance, especially in the presence of class imbalances, we compute standard and weighted accuracy, precision, recall, and the F1-score.

Regression For continuous targets, we evaluate brain age estimation using the healthy cohorts from Cam-CAN. During the training of Deep Learning models, these continuous targets are standardized. To prevent data leakage, the standardization parameters are fitted exclusively on the training set and subsequently applied to the validation and test sets. Model performance is quantified using a robust set of metrics to capture different aspects of the error distribution: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) (both standard and sample-weighted), R^2 score, explained variance, Mean Absolute Percentage Error (MAPE), and the Pearson correlation coefficient.

2.5 Evaluation protocols

To rigorously evaluate the predictive models, we employ a stratified k -fold cross-validation strategy with $k = 5$. This ensures fixed, reproducible train and test splits while maintaining **the original target distribution (e.g., class balance for classification or target range for regression)** across all folds. Strict separation between training and testing data is maintained at all stages; preprocessing steps, feature normalization, and hyperparameter tuning are fitted exclusively on the training set of each fold and subsequently applied to the held-out test set, preventing any information leakage. Model performance is aggregated across all folds and reported as the mean $\pm$ standard deviation for each evaluation metric.

3 Results

We systematically quantify the predictive utility of diffusion MRI microstructural representations across datasets, tasks, tissue compartments, and model families. Rather than focusing solely on peak accuracy, we decompose performance into configuration-specific effects and fold-level variability, enabling explicit assessment of stability, tissue-specific contributions, and the incremental value of diffusion-derived features beyond non-diffusion baselines.

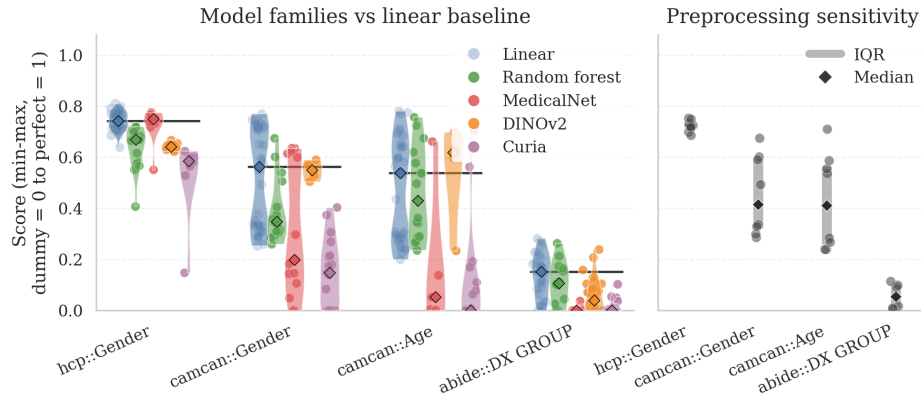


Fig. 2. Combined analysis of model-family performance and preprocessing sensitivity across four tasks using a normalized score (dummy baseline = 0, perfect performance = 1), enabling comparison across heterogeneous evaluation metrics. Left: Cross-validation performance of individual runs grouped by model family. Points denote fold-averaged scores per model; violins illustrate the distribution within each family; diamonds indicate medians. Horizontal lines mark the median performance of the linear baseline for each dataset–task. Right: Sensitivity to preprocessing strategy, aggregating across models and grouping by feature $\times$ tissue configurations. Each dot corresponds to a specific preprocessing strategy; vertical bars indicate the interquartile range (IQR) and overall spread across strategies. Variability induced by preprocessing is often comparable to, or larger than, differences between model families.

Figure 2 highlights two central findings. First, when evaluated on a common normalized scale, performance differences between model families are consistently modest. Regularized linear models remain competitive with higher-capacity architectures, and gains attributed to deep learning are strongly task-dependent. This suggests that representational choices may be as influential as architectural complexity.

Second, variability across preprocessing and microstructural parameterizations is substantial. In several tasks, the dispersion induced by feature $\times$ tissue configurations rivals or exceeds the differences observed between model families. This sensitivity indicates that predictive performance in diffusion MRI cannot be attributed to model choice alone. Instead, preprocessing strategy constitutes a major and often underreported source of variance. Consequently, conclusions drawn from a single representation or configuration may be unstable, underscoring the need for robustness analyses across preprocessing settings.

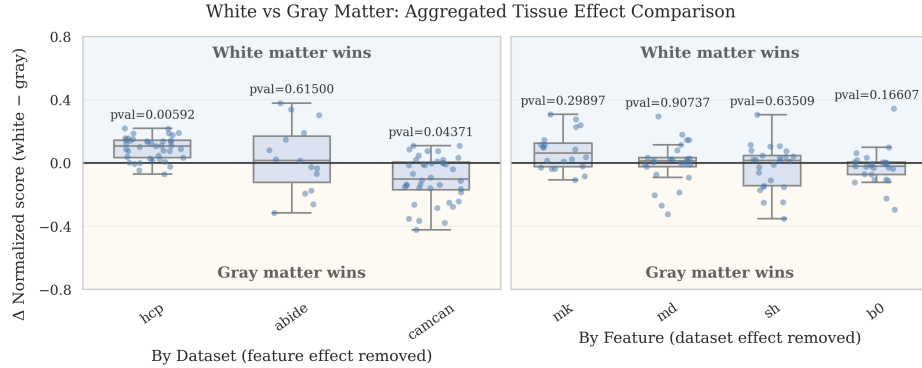


Fig. 3. Fold-level paired comparison of white versus gray matter performance using normalized scores (dummy baseline = 0, perfect performance = 1). Each point represents a matched fold-specific difference ($\Delta = \text{white} - \text{gray}$) under identical dataset-target-task-feature conditions. Left: dataset-level tissue effects after removing feature-specific baseline offsets. Right: feature-level tissue effects after removing dataset-task baseline offsets. Boxplots summarize the distribution of fold-level differences, and p-values are computed from configuration-level mean effects tested against zero. Positive values indicate white matter superiority; negative values indicate gray matter superiority.

We next quantify the impact of tissue compartment by performing fold-level paired comparisons between the best-performing white and gray matter models under identical dataset-target-task-feature configurations. For each fold, performance is normalized relative to a dummy baseline and the difference $\Delta = \text{white} - \text{gray}$ is computed, yielding a matched, configuration-specific tissue effect. To disentangle tissue influence from dataset and representation biases, we residualize the effects in two complementary ways: (i) removing feature-

specific baselines to isolate dataset-dependent tissue effects, and (ii) removing dataset–task baselines to isolate feature-dependent tissue effects. Statistical inference is performed on configuration-level mean effects to account for fold dependence, testing whether the tissue difference distribution deviates from zero.

As shown in Figure 3, tissue effects are strongly dataset-dependent. In HCP, we observe a systematic predominance of white matter over gray matter, with positive effect distributions indicating consistent white-matter superiority across configurations. In contrast, CamCAN exhibits the opposite pattern, with gray matter showing a stable advantage. For ABIDE, the distribution of differences is centered around zero, suggesting no clear predominance of either compartment.

When stratified by microstructural representation, most features do not demonstrate a consistent tissue preference. An exception is mean kurtosis (MK), which shows a modest but systematic white-matter advantage. For the remaining representations, tissue-driven differences are small and symmetrically distributed, indicating that their predictive contribution is largely compartment-agnostic. Importantly, the magnitude of these tissue effects remains comparable to cross-fold variability, underscoring that compartment selection constitutes a meaningful yet configuration-sensitive design choice. Collectively, these findings demonstrate that tissue preference is neither universal nor negligible, but instead emerges from the interaction between dataset characteristics and specific microstructural representations.

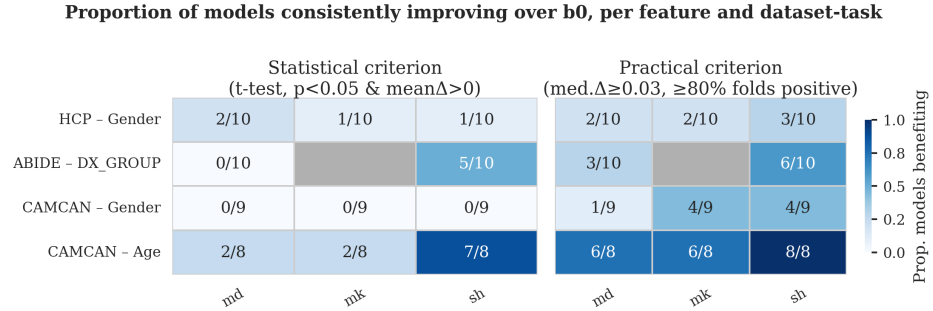


Fig. 4. Proportion of models exhibiting consistent performance improvement over the non-diffusion baseline (b_0), stratified by dataset–task and microstructural feature. Improvement is evaluated relative to fold-level normalized scores ($\Delta = \text{feature} - b_0$). Left: statistical criterion (one-sample t-test against zero on fold-level deltas; $p < 0.05$ and $\text{mean } \Delta > 0$). Right: robustness criterion (median $\Delta \geq 0.03$ and 80% of folds positive). Each cell reports the fraction of models satisfying the corresponding criterion (numerator/total models). Rows correspond to dataset–target combinations; columns correspond to diffusion-derived features.

To assess whether diffusion-derived microstructural features provide predictive value beyond non-diffusion signal, we quantified model-wise improvements relative to b_0 . The results reveal that diffusion information does not uniformly

enhance performance across tasks or datasets. For proxy tasks such as gender classification in HCP and CamCAN, diffusion features do not confer a systematic advantage over b_0 , indicating that non-diffusion signal (likely reflecting global anatomical properties such as brain size) already captures most of the discriminative information. In contrast, tasks more directly linked to microstructural variation show measurable gains from diffusion metrics. Age prediction in CamCAN and diagnostic classification (DX_GROUP) in ABIDE exhibit consistent improvements for several diffusion representations, particularly under linear models operating on array-based features (e.g., spherical harmonics).

Notably, deep learning models tend to exhibit positive gains in certain gender classification settings, likely reflecting their sensitivity to global morphological cues embedded in image-based representations rather than diffusion-specific contrast. Overall, these findings indicate that the added value of diffusion MRI depends strongly on the biological relevance of the task, the dataset characteristics, and the modeling framework, rather than representing a universal performance boost.

4 Conclusion

We introduced a standardized diffusion MRI benchmark for demographics prediction in individuals across three large-scale datasets: Human Connectome Project, Cambridge Centre for Ageing and Neuroscience, and Autism Brain Imaging Data Exchange. Within a unified experimental framework, we systematically compared multiple microstructural representations (b_0 , MD, SH, MK) and model families spanning from dummy baselines to linear, ensemble, and deep learning models. By controlling preprocessing, model selection, and evaluation procedures, the benchmark enables fair comparison across configurations and substantially reduces methodological heterogeneity that currently limits reproducibility in diffusion MRI prediction studies.

Our results reveal that predictive performance in diffusion MRI is highly configuration-dependent. Increased model complexity does not reliably translate into superior performance: well-regularized linear models frequently match or outperform more sophisticated alternatives. More importantly, variability across folds, tissue compartments, and feature representations often rivals mean performance differences. Tissue preference is dataset-specific (e.g., white matter predominance in HCP, gray matter predominance in CamCAN, no clear advantage in ABIDE), and most microstructural features do not exhibit universal superiority. Diffusion-derived metrics provide measurable gains only in tasks closely aligned with microstructural variation, whereas for proxy tasks, non-diffusion signal can be equally informative.

Collectively, these findings challenge the assumption that diffusion features or complex models inherently yield robust improvements. Instead, they indicate that many reported gains may reflect sensitivity to specific configurations rather than stable, generalizable signal extraction. By explicitly decomposing performance into configuration-specific effects and variability components, this

benchmark reframes diffusion MRI prediction as a problem of stability, representational validity, and controlled comparison.

This work deliberately prioritizes controlled within-dataset evaluation to establish internal robustness under standardized conditions. Future extensions should assess cross-cohort transferability, broader prediction targets, and alternative preprocessing strategies to determine whether observed variability is amplified or mitigated under distributional shift.

Ultimately, we hope this framework encourages a shift from isolated, configuration-specific reporting toward reproducible, variance-aware analysis, and transparent methodological reporting in diffusion MRI. Establishing shared benchmarks and open implementations is a necessary step toward methodological unification and more reliable neuroimaging-based prediction.

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